

Con-Fog: Consensus-Driven Fog Node Selection in FU-Serve Platform for IoT Applications

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Abstract—The rapid expansion of Internet of Things (IoT) devices and applications necessitates the need for more efficient computational and data management strategies. The fog-enabled UAV-as-a-Service (FU-Serve) platform addresses these demands by integrating fog computing to enhance the operational efficiency of UAVs in IoT environments. Despite its advantages, the FU-Serve platform faces significant challenges, including data transmission latency, resource allocation, and energy management, contributing to the underutilization of UAVs and fog nodes. To address these challenges, this article introduces a consensus-driven approach, Con-Fog, that optimizes the selection of fog nodes for UAVs within the FU-Serve platform. Con-Fog evaluates potential fog nodes within the communication range by computing utility values based on geographical distance, link quality, available computational resources, and residual energy. UAVs rank these nodes according to their utility values and select the most suitable ones through a consensus-based approach, ensuring alignment with the operational demands of IoT devices. Additionally, we apply an optimal best-fit algorithm to refine fog node allocation, maximizing resource utilization while keeping it below each node's capacity threshold ($\mathcal{T}\%$). Our simulation results show that Con-Fog significantly enhances key IoT performance metrics. Transmission time and the number of unassigned UAVs decrease by 10%–30% and 20%–40%, respectively, while residual energy increases by 30%–50% compared to existing systems. These improvements enhance the management of UAV and fog node resources, thereby advancing the effectiveness of IoT applications within the FU-Serve platform.

Index Terms—Fog computing, Internet of Things (IoT), social choice theory, UAV-as-a-Service (UAVaaS), Unmanned Aerial Vehicles (UAVs).

I. INTRODUCTION

IN THE evolving landscape of Unmanned Aerial Vehicles (UAVs), their integration with the Internet of Things (IoT) has transformed multiple sectors. This integration significantly enhances UAV operations across diverse fields, including agriculture, transportation, surveillance, smart cities, structural

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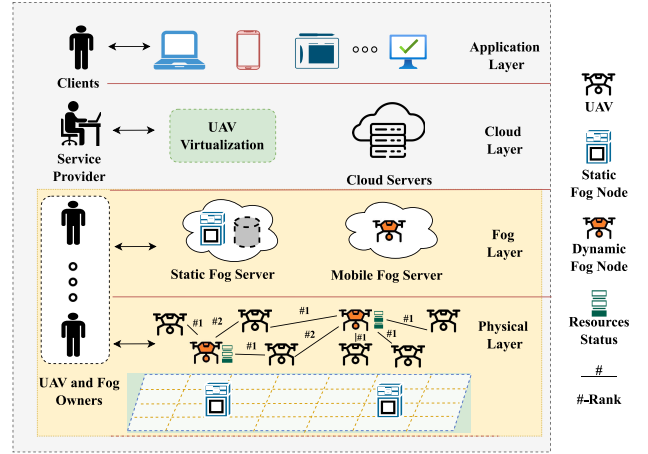


Fig. 1. FU-serve platform.

inspections, disaster response, and military applications [1], [2], [3], [4], [5], [6], [7]. Despite the broad capabilities of UAVs, traditional UAV operations have been designed for single-user applications requiring end-users to handle deployment, operation, and management. These responsibilities often present significant challenges to end-users [8]. To address these challenges, Yapp et al. [9] introduced the concept of UAV-as-a-Service (UAVaaS), based on the similar principles of the traditional cloud computing service models. However, the UAVaaS platform's reliance on cloud technology for back-end processing introduces significant latency. This latency poses challenges for applications requiring immediate decision-making, emphasizing the need for more robust solutions. Addressing this critical need, Raju and Roy [10] developed a Fog-enabled UAVaaS platform (FU-Serve), which is tailored to enhance the responsiveness of time-critical applications. By employing fog computing, the FU-Serve platform brings computational tasks closer to the source than traditional cloud architectures. This proximity allows the platform to efficiently segregate UAV-sensed data into time-critical and nontime-critical categories. Time-critical data is processed directly at the fog layer, which significantly improves network response times [11]. In contrast, non time-critical data are processed and stored in the traditional cloud. This bifurcation of data efficiently handles data by ensuring that each type of data is processed most efficiently.

A FU-Serve platform consists of two types of fog nodes—static and dynamic—as depicted in Fig. 1. A static fog node exists at a consistent location and is equipped with finite computational capacity, bandwidth, and an uninterrupted power

supply, making them ideal for predictable environments [11]. On the other hand, a dynamic fog node, essentially UAVs designated for specific operational zones, offers mobility as its key feature [12], suited for deployment in more volatile or hostile environments [13]. Each type of fog node brings distinct strengths and challenges to the FU-Serve platform. Data sensing and processing within the FU-Serve platform rely heavily on the interaction between UAVs and fog nodes. When UAVs detect time-critical data, they send it to the nearest fog node for immediate response. However, selecting a fog node based solely on distance leads to inefficiencies [14]. This method overlooks critical factors such as the current workload and computational capabilities of the fog nodes. As a result, it causes delays and extended data processing times, particularly during peak usage or when several UAVs concurrently transmit data to the same node.

Moreover, Uneven UAV allocation to fog nodes often leads to an imbalanced load sharing among the UAVs. Consequently, some nodes become overloaded while others remain under-utilized. Therefore, designing a suitable UAV assignment strategy with a fog node is required to enhance efficient resource allocation. Considering such a situation, we introduce a novel fog node selection strategy that is based on social choice theory for optimal resource distribution [15], [16]. In this approach, we consider each UAV as a member of a society, collectively assessing the utility of potential fog nodes. UAVs calculate each fog node's utility values based on resource capacities, operational demands, geographical proximity, link quality, and residual energy levels. Following this evaluation, the UAVs employ a consensus-based mechanism derived from social choice theory to collectively rank the fog nodes. Such a consensus mechanism ensures that the decision-making process accounts for the preferences of all UAVs, reflecting a democratic approach where the collective ranking of nodes leads to the selection of the most suitable fog nodes. The proposed approach, Con-Fog, significantly enhances the performance of the FU-Serve platform while balancing load distribution and extending UAV operational longevity. It outperforms traditional UAVaaS platforms that rely on centralized cloud processing, which often suffer from high latency and scalability constraints [9]. Moreover, Con-Fog refines the initial proximity-based fog node selection approach of the FU-Serve platform, which previously led to suboptimal resource utilization [10]. This optimization results in markedly improved resource management.

A. Motivation

The FU-Serve platform utilizes three types of UAVs, including Fog-UAVs (F-UAVs), Hybrid UAVs (H-UAVs), and Normal-UAVs (N-UAVs), while each serves distinct roles within the IoT ecosystem [10]. F-UAVs, equipped with advanced fog services but lacking sensing capabilities, and H-UAVs, which combine sensing with limited fog computing powers, are pivotal in managing emergency responses. Traditionally, fog node selection focused on proximity, often overlooking the complex operational demands and diverse resource capacities critical for effective IoT resource allocation. Consequently, nearby fog nodes can become overutilized,

leading to operational inefficiencies. On the other hand, FU-Serve is a unique and recently proposed paradigm. Therefore, the existing fog node selection approach is not suitable for FU-Serve platform. This oversight motivates us to design a fog node selection specifically for the FU-Serve platform while promoting balanced fog node use and ensuring efficient mission execution driven by IoT.

B. Contributions

We introduce a novel fog node selection mechanism while applying the social choice theory. The specific contributions of this work are as follows.

- 1) In order to select a suitable fog node, we map the problem with a social-choice theory. We consider that each of the UAVs is forming a society where they compute the utility value of each potential fog node. Further, the UAVs apply the consensus-based mechanism to rank these fog nodes based on the calculated utility values, employing a consensus-driven approach to select the most suitable nodes that meet their operational demands.
- 2) We apply an optimal best-fit algorithm to enhance resource allocation and extend the operational lifespan of fog nodes. This algorithm efficiently distributes UAVs across fog nodes, maximizing resource utilization while maintaining usage below a predefined threshold for each node. Such a resource allocation ensures that a minimal number of fog nodes are used to their maximum capacity, preserving additional nodes for future allocations, which improves the system's sustainability and efficiency.
- 3) The FU-Serve platform is new and different from the existing service-oriented architecture. Moreover, the proposed approach, Con-Fog, is specifically designed for the FU-Serve platform. Therefore, we evaluate the performance of Con-Fog with the help of rigorous simulation.

II. RELATED WORK

In this section, we highlight the literature those are related to our work. We divided the literature survey into two sections:

- 1) UAVaaS and 2) Fog node selection.

A. UAV-as-Service

The UAVs are widely used devices in different applications such as providing cost-effective data collection, enhanced safety, precision agriculture, and infrastructure inspection [17]. The use of traditional UAVs is based on single-user centricity, where an owner procures and uses the UAVs for serving his/her application. However, Yapp et al. [9] designed a UAVaaS paradigm, which addresses the issue of single-user centricity. UAVaaS platform considers the scenarios where a single UAV needs to serve multiple clients at once, a critical functionality for maximizing resource efficiency in constrained environments. Expanding on this concept, Alkouz et al. [18] developed a swarm-based drone-as-a-service (SDaaS) targeting the delivery operations, concentrating on drone swarm strategies to enhance delivery efficiency while managing battery and recharging limitations. The authors used modified

A* algorithms and cooperative behaviors, demonstrating the significant capabilities of drone swarms in improving delivery systems. Additionally, Pathak et al. [19] explored the architecture of UAVaaS, including the virtualization of UAVs to enable efficient service distribution through social welfare-based grouping and virtualization strategies.

To optimize UAV selection within UAVaaS, Roy and Bouvry [20] designed an approach for UAV selection while considering their capabilities and usage patterns, aiming to deliver high-quality services and address the constraints of UAV flight times. In spite of this method improving service optimization, it is unable to handle time-sensitive applications due to its reliance on centralized processing. Building on these insights, Raju and Roy [10] advanced the UAVaaS model by incorporating a fog computing layer, significantly reducing latency and enhancing real-time data transfer capabilities for IoT applications. The adaptation involves selecting an optimal fog node for data processing of the time-critical applications and optimizing the overall system efficiency.

B. Fog Node Selection Approaches

A significant amount of work has already been explored in the existing literature on fog computing. Alam et al. [21] employed machine learning to predict the movements of mobile nodes to align these predictions with fog nodes' service regions for streamlining task offloading, focusing on Quality of Service (QoS) and resource availability. However, their method did not consider the mobile fog nodes. Similarly, Zhang et al. [22] developed a distributed optimization framework using a Stackelberg game for resource allocation using many-to-many matching games, which did not consider the dynamic nature of fog nodes, limiting its effectiveness in mobile environments. Additionally, Misra and Chatterjee [23] approached multihop task offloading with a heuristic-driven integer linear program that considered latency and energy efficiency, but the proposed work is suitable for static networks rather than mobile networks.

The selection of a suitable fog node is a crucial task in efficiently executing a system [24]. Therefore, many authors proposed different approaches to select a suitable fog node in a fog-based system. Baranwal and Vidyarthi [25] proposed the decentralized Fog Orchestrator Node Selection (FONS) model, enhancing the flexibility and reliability of application placement, marking a significant improvement in fog node selection despite recognizing the need for further research into fog node dynamics at physical layer. Similarly, Jie et al. [26] and Ogundoyin and Kamil [27] introduced novel fog node allocation methods based on game-theoretic and multiobjective optimization strategies, respectively, both aimed at optimizing resource utilization in dynamically changing fog computing environments. Rahman et al. [28] explored the dynamic reconfiguration of edge nodes to handle multiple IoT sensor requests efficiently. Their approach enhances the utilization of the network and reduces delays by employing different methods such as SELF, LCBF, and RS. However, this approach does not address the computational needs essential for minimizing latency and energy use. Additionally, it falls

short of effectively handling diverse service demands. On the other hand, Karthikeyan and Brindha [29] presented a framework that leverages Ant Colony Optimization for improving fog node selection. This approach aims to optimize resource allocation by considering factors such as node capacity, security, and availability. While the proposed approach significantly improves efficiency and responsiveness, it lacks the capability to make real-time adjustments. The framework does not incorporate continuous network performance data, which is crucial for dynamically optimizing resource allocation in response to changing network conditions and varying workload demands. Lv et al. [30] proposed a game-theoretic approach to optimize fog node selection for data offloading. The authors used a Stackelberg game-theoretic approach to formulate the interactions between clients and fog nodes. Using the Nash equilibrium strategies, the authors demonstrated effective reductions in offloading delay and improved fog node revenue. However, the study does not fully explore adaptive mechanisms that could dynamically recalibrate fog node selection based on ongoing changes in network conditions. Conversely, Bashir et al. [31], Misra and Saha [32], and Marbukh [33] designed a mechanism focused on improving load assessments, data aggregation in critical health scenarios, and balancing service preferences with energy conservation. Similarly, Singh et al. [34] developed a method that optimizes task allocation among fog nodes using binary particle swarm optimization to minimize task execution and transfer times. While this method significantly improves load distribution and computational efficiency in fog networks, it does not fully explore the potential for automated reconfiguration based on real-time analytics of node performance, which could dynamically adapt to changes without predefined thresholds.

Synthesis: The existing literature on UAVaaS and fog node selection aims to enhance UAV services and improve operational efficiencies using the concept of virtualization and node selection strategies. We notice that the literature focuses on advancements in reducing latency, improving data processing, and optimizing resource allocation within dynamic environments. However, despite these developments, the existing mechanisms struggle with real-time decision-making in dynamically changing conditions. Specifically, many fog node selection strategies are not suitable for the UAVaaS platform due to the presence of the unique operational demands and energy constraints of UAVs. Therefore, we witness a critical gap—the need for fog node selection frameworks tailored to UAV selection for the UAVaaS platform.

III. PROBLEM STATEMENT

Within the FU-Serve platform, we address the challenge of optimally pairing UAVs with fog nodes.

A. Scenario Description

We consider a set of UAVs $U = \{u_0, u_1, \dots, u_m\}$, and a set of fog nodes, represented as $F = \{f_0, f_1, \dots, f_n\}$, deployed over an application region. The primary goal is to assign each UAV to an optimal fog node that maximizes computational

efficiency, minimizes communication costs, and conserves energy, enhancing overall system performance.

B. Problem Formulation

The objective is to select a fog node f_j for each UAV u_i such that the operational demands are efficiently met for serving the applications of a u_i . The optimization problem is defined by the function $\phi : U \rightarrow F \cup \{\text{unassigned}\}$, which assigns each UAV to a fog node or marks it as unassigned. The cost function minimized is described by

$$\arg \min_c \mathcal{Z} = \sum_{i=0}^m \sum_{j=0}^n c(u_i, f_j) \cdot x_{ij} + \sum_{i=0}^m p \cdot \left(1 - \sum_{j=0}^n x_{ij} \right) - \sum_{j=0}^n \mathcal{E}_j^{f, \text{res}} \cdot x_{ij} \quad (1)$$

where $c(u_i, f_j)$ denotes the communication cost, p is the penalty for unassigned UAVs, $\mathcal{E}_j^{f, \text{res}}$ represents the residual energy of fog node f_j , and x_{ij} is a binary decision variable indicating the assignment of UAV u_i to fog node f_j

$$x_{ij} = \begin{cases} 1, & \text{if } u_i \text{ is assigned to } f_j \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

C. Constraints

The optimization is subject to constraints that ensure effective and efficient system operation:

Assignment Constraint: Each UAV must be assigned to exactly one fog node. This is formulated as

$$\sum_{j=0}^n x_{ij} = 1 \quad \forall i \in \{0, \dots, m\}. \quad (3)$$

Resource Capacity Constraint: The total demand placed on any fog node by the assigned UAVs must not exceed its resource capacity, described by

$$\sum_{i=0}^m \mathcal{R}_i^{u, \text{req}} \cdot x_{ij} \leq \mathcal{T} \cdot \mathcal{R}_j^{f, \text{cur}} \quad \forall j \in \{0, \dots, n\} \quad (4)$$

where $\mathcal{R}_i^{u, \text{req}}$ denotes the resource requirements of UAV u_i and $\mathcal{R}_j^{f, \text{cur}}$ denotes the current available resources of fog node f_j . $\mathcal{T}$ is the threshold factor that lies between $0 < \mathcal{T} \leq 100$, dictating the maximum allowable percentage of resource utilization at each fog node.

Energy Constraint: The energy resources of the fog nodes should not be depleted by the demands of the UAVs beyond their capacity

$$\mathcal{E}_j^{f, \text{max}} \geq \sum_{i=0}^m \mathcal{E}_i^{u, \text{req}} \cdot x_{ij} \quad \forall j \in \{0, \dots, n\}. \quad (5)$$

Proximity Constraint: This constraint aims to minimize the physical distances to reduce transmission latency and energy consumption

$$\min \sum_{i=0}^m \sum_{j=0}^n d(u_i, f_j) \cdot x_{ij}. \quad (6)$$

TABLE I
MATHEMATICAL NOTATION AND DESCRIPTION

Symbol	Description
U	Set of UAVs $U = \{u_0, \dots, u_m\}$
F	Set of fog nodes $F = \{f_0, \dots, f_n\}$
ϕ	Function mapping UAVs to fog nodes
$c(u_i, f_j)$	Cost of assigning UAV u_i to fog node f_j
x_{ij}	Binary variable for assignment of UAV u_i to fog node f_j
$\mathcal{E}_j^{f, \text{max}}$	Max energy of fog node f_j
$\mathcal{E}_u^{f, \text{req}}$	Energy requirements of UAV u_i
$\mathcal{E}_j^{f, \text{res}}$	Residual energy of fog node f_j
$\mathcal{R}_j^{f, \text{max}}$	Max resources at fog node f_j
$\mathcal{R}_i^{u, \text{req}}$	Resource requirements of UAV u_i
$\mathcal{R}_j^{f, \text{cur}}$	Available resources at fog node f_j
$\mathcal{T}$	Threshold for maximum resource utilization at fog nodes
$d(u_i, f_j)$	Distance between UAV u_i and fog node f_j
$\mathcal{C}_{u,i}^r$	Communication range of UAV u_i
$\chi_{\mathcal{C}_{u,i}^r}(f_j)$	Characteristic function for range compliance of fog node f_j
$\mathcal{D}_{u_i, f_j}^{\text{eff}}$	Effective distance between UAV u_i and fog node f_j
$\mathcal{R}_{f_j}^{\text{eff}}$	Effective resource availability at fog node f_j
$\mathcal{L}^{\text{eff}}(u_i, f_j)$	Effective link quality index between UAV u_i and fog node f_j
$\mathcal{E}_{f_j}^{\text{eff}}$	Effective residual energy at fog node f_j
$\mathcal{U}_{i,j}$	Utility function for UAV u_i selecting fog node f_j , combining effective distance, resource availability, link quality, and energy efficiency
$\mathcal{C}_{f,j}^s$	Consensus-Based Average Preference Score (CAP Score), $\mathcal{C}_{f,j}^s$, for each fog node f_j
$(\alpha, \beta, \gamma, \delta)$	Weights in the utility function reflecting the importance of each parameter

These constraints ensure that the system's operational efficiency is maintained by balancing the resource and energy demands across the network, thereby facilitating optimal UAV-to-fog node assignments within the FU-Serve platform. The detailed mathematical notations and their description are shown in Table I.

IV. FOG NODE SELECTION USING CON-FOG APPROACH

In the FU-Serve platform, assigning UAVs to the most suitable fog nodes is essential for minimizing latency and maximizing resource utilization. This assignment is governed by a comprehensive decision-making process that rigorously assesses the capabilities of each fog node. The process emphasizes evaluating the CPU cycles, memory, and storage capacities of the fog nodes to ensure they meet the specific computational requirements of the UAVs. This strategic assignment is essential for optimizing the overall efficiency and performance of the FU-Serve platform.

A. Consensus-Driven Fog Node Selection

In our approach, we implement a consensus-driven decision-making process based on social choice theory to select the most suitable fog nodes [35]. This approach integrates individual UAV preferences into a collective framework, thoroughly

assessing each fog node across multiple critical dimensions: computational resources, link quality, and residual energy levels. We calculate the effective distance, resource availability, link quality index, and residual energy for each fog node to ensure a comprehensive evaluation. This detailed assessment supports optimal UAV-to-fog node assignments, enhancing the efficiency and performance of the FU-Serve platform.

Effective Distance ($\mathcal{D}^{\text{eff}}$): Distance between a UAV u_i and a fog node f_j within its range $\mathcal{C}_{u,i}^r$, computed as

$$\mathcal{D}_{u_i,f_j}^{\text{eff}} = \frac{\sqrt{(x_{u_i} - x_{f_j})^2 + (y_{u_i} - y_{f_j})^2}}{\sum_{k=1}^n \sqrt{(x_{u_i} - x_{f_k})^2 + (y_{u_i} - y_{f_k})^2}} \cdot \chi_{\mathcal{C}_{u,i}^r}(f_j) \quad (7)$$

where x_{u_i} and y_{u_i} are the coordinates of UAV u_i , x_{f_j} and y_{f_j} are the coordinates of fog node f_j , and $\chi_{\mathcal{C}_{u,i}^r}(f_j)$ is the characteristic function, which equals 1 if the fog node f_j is within the communication range $\mathcal{C}_{u,i}^r$ of the UAV u_i , and 0 otherwise.

Effective Resource Availability ($\mathcal{R}^{\text{eff}}$): Indicates the availability of resources at fog node f_j as

$$\mathcal{R}_{f_j}^{\text{eff}} = \frac{\mathcal{R}_j^{f,\text{cur}}}{\mathcal{R}_i^{u,\text{req}} \cdot \sum_{k=1}^n \mathcal{R}_k^{f,\text{cur}}} \quad (8)$$

Effective Link Quality Index ($\mathcal{L}^{\text{eff}}$): Reflects the link quality between UAV u_i and fog node f_j , computed as

$$\mathcal{L}^{\text{eff}}(u_i, f_j) = \frac{\mathcal{L}(u_i, f_j)}{\sum_{k=1}^n \mathcal{L}(u_i, f_k)} \quad (9)$$

where $\mathcal{L}(u_i, f_j)$ is the link quality indicator (LQI) between UAV u_i and fog node f_j , primarily determined by the signal-to-noise ratio (SNR)

$$\mathcal{L}(u_i, f_j) = \text{SNR}_{u_i,f_j} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (10)$$

Here, P_{signal} is the power of the received signal at fog node f_j from UAV u_i , and P_{noise} is the power of the background noise.

Effective Residual Energy ($\mathcal{E}^{\text{eff}}$): Represents the residual energy of fog node f_j

$$\mathcal{E}_{f_j}^{\text{eff}} = \frac{\mathcal{E}_j^{f,\text{res}}}{\sum_{k=1}^n \mathcal{E}_k^{k,\text{res}}} \quad (11)$$

These parameters are meticulously chosen to provide a comprehensive overview of both the operational and environmental conditions influencing each fog node, thereby facilitating a well-informed and balanced UAV-to-fog node assignment process. For instance, we compute $\mathcal{D}^{\text{eff}}$ while considering the proximity of all potential fog nodes. $\mathcal{D}^{\text{eff}}$ significantly reduces latency and energy consumption, factors that are crucial for an IoT operations. Additionally, $\mathcal{R}^{\text{eff}}$ ensures that resources are equitably distributed across fog nodes, while prevents any single node from becoming a bottleneck, thereby enhancing the overall network reliability. We compute $\mathcal{L}^{\text{eff}}$ to measure the robustness of the communication link, deliberately prioritizing UAV connections to fog nodes that demonstrate optimal performance, crucial for maintaining data integrity. On the other hand, $\mathcal{E}^{\text{eff}}$ determines the energy profiles of fog nodes, aiming to support extended operational periods

without the need for frequent recharging—a vital attribute for continuous field operations.

The normalization and dimensionless characteristics of these parameters facilitate their seamless incorporation into the utility function, which aggregates multiple criteria into a unified score for each fog node. This approach aligns with the multicriteria decision-making frameworks outlined in [22], [23], [24], [29], and [30], where normalized parameters have been demonstrated to enhance the fairness and precision of resource allocation decisions.

Upon the individual assessments of effective distance, resource availability, link quality, and residual energy, we define a utility function for each UAV u_i in selecting a fog node f_j . This utility function, shown in (12), synthesizes these parameters to optimize resource use, link quality, and energy efficiency

$$\mathcal{U}_{i,j} = \frac{\alpha}{\mathcal{D}_{u_i,f_j}^{\text{eff}-1}} + \beta \cdot \mathcal{R}_{f_j}^{\text{eff}} + \gamma \cdot \mathcal{L}^{\text{eff}}(u_i, f_j) - \delta \cdot \mathcal{E}_{f_j}^{\text{eff}}. \quad (12)$$

The weights α, β, γ , and δ reflect the importance of each parameter, tailored to the operational needs of UAVs. In our proposed approach, each UAV identifies suitable fog nodes based on maximum utility values, subject to the constraints depicted in (3)–(6). Therefore, we have

$$\arg \max_{i \leq m, j \leq n} \mathcal{U}_{i,j}. \quad (13)$$

Each UAV forms an opinion based on its utility value, represented as a Utility matrix

$$\mathcal{M} = \begin{bmatrix} \mathcal{U}_{1,1} & \cdots & \mathcal{U}_{1,n} \\ \vdots & \ddots & \vdots \\ \mathcal{U}_{m,1} & \cdots & \mathcal{U}_{m,n} \end{bmatrix} \quad (14)$$

where $\mathcal{U}_{i,j}$ is the utility value between the i^{th} UAV and j^{th} fog node. Based on these utility values, each UAV ranks the available fog nodes, identifying the most suitable one for its needs. This ranking is captured in

$$R = \begin{bmatrix} r_{1,1} & \cdots & r_{1,n} \\ \vdots & \ddots & \vdots \\ r_{m,1} & \cdots & r_{m,n} \end{bmatrix}. \quad (15)$$

Here, $r_{i,j}$ indicates the rank assigned by UAV i to fog node j , with lower values showing a higher preference.

While individual rankings provide insights into each UAV's preferences, relying solely on them leads to imperfect global decisions. For instance, some fog nodes might be highly preferred by many UAVs, leading to overutilization, while others remain underutilized. To address this issue and ensure a fair and balanced allocation of UAVs across fog nodes, we introduce the consensus-based average Preference Score (CAP Score), $\mathcal{C}_{f,j}^s$. The CAP Score, $\mathcal{C}_{f,j}^s$, integrates all UAV preferences into a single metric that reflects the collective opinion of the UAVs about each fog node. It is calculated as

$$\mathcal{C}_{f,j}^s = \frac{\sum_{i=1}^m r_{i,j}}{|\{i \in U : r_{i,j} > 0\}|} \quad (16)$$

Algorithm 1 Consensus-Based Fog Node Selection**Require:**

Set of UAVs $U = \{u_1, u_2, \dots, u_m\}$
 Set of fog nodes $F = \{f_1, f_2, \dots, f_n\}$
 Communication Range $C_{u,i}^r$ of each u_i

Ensure: Optimal Assignment of UAVs to Fog Nodes

```

1: for each  $u_i$  in  $U$  do
2:   for each  $f_j$  in  $F$  do
3:     if  $f_j$  is within  $C_{u,i}^r$  of  $u_i$  and has required  $\mathcal{R}_j^f$  then
4:       FognodesInRange[ $u_i$ ].add( $f_j$ )
5:     end if
6:   end for
7: end for
8: for each  $u_i$  in  $U$  do
9:   for each  $f_j$  in FognodesInRange[ $u_i$ ] do
10:    Calculate  $\mathcal{U}_{i,j}$  using the utility equation (12)
11:    if  $\mathcal{U}_{i,j}$  is higher than previous  $\mathcal{U}_{i,j}$  then
12:      Update  $\mathcal{U}_{i,j}$  in the  $\mathcal{M}$ 
13:    end if
14:   end for
15: end for
16: for each  $u_i$  in  $U$  do
17:   Sort FognodesInRange[ $u_i$ ] by  $\mathcal{U}_{i,j}$  in descending order
18:   Assign ranks to fog nodes in FognodesInRange[ $u_i$ ]
    based on sorted order
19: end for
20: for each  $f_j$  in  $F$  do
21:   Calculate  $C_{f,j}^s$  using the CAP Score equation (16)
22: end for
23: Sort  $F$  by  $C_{f,j}^s$  in ascending order
24: Execute Algorithm 2
25: return Assignment of UAVs to Fog Nodes

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This score normalizes the average ranking for each fog node by the number of UAVs that have ranked it, ensuring a fair assessment across all fog nodes. By aggregating individual preferences into a consensus-based score, the CAP Score mitigates the risk of overloading specific fog nodes and ensures equitable resource distribution.

The fog nodes are then ordered by their CAP scores for optimal assignment

$$\mathcal{S}^f = \sigma(\{f_1, f_2, \dots, f_n\}) \quad (17)$$

where σ is a permutation of indices $\{1, 2, \dots, n\}$ such that

$$C_{f,\sigma(1)}^s \leq C_{f,\sigma(2)}^s \leq \dots \leq C_{f,\sigma(n)}^s \quad (18)$$

and $C_{f,\sigma(j)}^s$ denotes the CAP Score of fog node f_j . The permutation σ sorts the nodes from those with the lowest to the highest CAP score, facilitating an efficient and equitable fog node selection process. This structured arrangement supports decision-making and resource allocation within the FU-Serve platform, ensuring that UAVs are assigned to fog nodes in a manner that balances load distribution, minimizes resource wastage, and enhances overall system performance. The detailed pseudocode for the consensus-based fog node selection is outlined in Algorithm 1.

Algorithm 2 Optimum Fog Node Resource Utilization**Require:**

Set of UAVs $U = \{u_1, u_2, \dots, u_m\}$
 Set of sorted fog nodes $F = \{f_1, f_2, \dots, f_n\}$ (sorted by CAP Scores $C_{f,j}^s$)
 Resource requirements $\mathcal{R}_i^{u,req}$ of each u_i
 Maximum resource capacity $\mathcal{R}_j^{f,max}$ of each f_j
 Currently available resources $\mathcal{R}_j^{f,cur}$ at each fog node f_j

Ensure: Assignment of UAVs to fog nodes

```

1: Initialize  $\mathcal{R}_j^{f,cur} \leftarrow 0$  for each fog node  $f_j$ 
2: for each fog node  $f_j$  in  $F$  do
3:   UAVList  $\leftarrow \{u_i \in U : f_j \text{ is ranked first by } u_i\}$ 
4:   Sort UAVList by  $\mathcal{R}_i^{u,req}$  in ascending order
5:   for each  $u_i$  in UAVList do
6:     if  $\mathcal{R}_j^{f,cur} + \mathcal{R}_i^{u,req} \leq \mathcal{T} \times \mathcal{R}_j^{f,max}$  then
7:       Assign  $u_i$  to  $f_j$ 
8:        $\mathcal{R}_j^{f,cur} \leftarrow \mathcal{R}_j^{f,cur} + \mathcal{R}_i^{u,req}$ 
9:       Remove  $u_i$  from  $U$ 
10:    end if
11:   end for
12: end for
13: return Assignment of UAVs to fog nodes

```

B. Optimum Fog Node Resource Allocation

After computing the rank of fog nodes based on their CAP Scores, Algorithm 2 assigns UAVs to fog nodes in a manner that maximizes resource utilization while ensuring no fog node exceeds its designated capacity threshold $\mathcal{T}\%$. We map the resource utilization problem as a bin-packing problem, where fog nodes represent bins with specific capacity constraints such that

$$\mathcal{R}_j^{f,cur} + \sum_{i=1}^m \mathcal{R}_i^{u,req} x_{ij} \leq \mathcal{T} \cdot \mathcal{R}_j^{f,max} \quad \forall j \in \{1, \dots, n\} \quad (19)$$

where $x_{ij} \in \{0, 1\}$ indicates whether UAV u_i is assigned to fog node f_j , $\mathcal{R}_i^{u,req}$ denotes the resource requirement of UAV u_i , and $\mathcal{R}_j^{f,max}$ represents the maximum resource capacity of fog node f_j .

The algorithm employs a greedy strategy, assigning UAVs to fog nodes in ascending order of their CAP Scores ($C_{f,j}^s$), ensuring that high-priority fog nodes are utilized first. Let $\mathcal{S}_j$ denotes the set of UAVs assigned to fog node f_j . The utility function $g(\mathcal{S}_j)$ for each fog node is defined as

$$g(\mathcal{S}_j) = \min \left(\mathcal{T} \cdot \mathcal{R}_j^{f,max}, \sum_{u_i \in \mathcal{S}_j} \mathcal{R}_i^{u,req} \right) \quad (20)$$

where $g(\mathcal{S}_j)$ represents the total resource utilization of fog node f_j , capped at its capacity threshold $\mathcal{T} \cdot \mathcal{R}_j^{f,max}$. This utility function ensures that the fog node does not exceed its designated capacity while maximizing the allocation of UAVs.

The function $g(\mathcal{S}_j)$ is submodular due to diminishing returns as $\mathcal{R}_j^{f,cur}$ approaches $\mathcal{T} \cdot \mathcal{R}_j^{f,max}$. Specifically, the marginal gain in resource utilization decreases as more UAVs are assigned to the fog node, reflecting the principle of diminishing returns.

By leveraging submodular optimization principles, the greedy strategy achieves at least $(1 - 1/e)$ of the optimal value for $g(S_j)$, ensuring near-optimal resource allocation.

Furthermore, we rigorously prove the algorithm's optimality using Theorem 2. Specifically, we demonstrate that the total unused resources across all fog nodes

$$\mathcal{U}_{\text{total}} = \sum_{j=1}^n (\mathcal{T} \cdot \mathcal{R}_j^{f,\max} - \mathcal{R}_j^{f,\text{cur}}) \quad (21)$$

is minimized. Any deviation from this strategy either violates the resource constraints

$$\sum_{i=1}^m \mathcal{R}_i^{u,\text{req}} x_{ij} > \mathcal{T} \cdot \mathcal{R}_j^{f,\max} \quad (22)$$

or results in higher unused resources ($\mathcal{U}'_{\text{total}} > \mathcal{U}_{\text{total}}$).

Additionally, Algorithm 2 incorporates a sorting mechanism for UAVs based on their resource requirements ($\mathcal{R}_i^{u,\text{req}}$). Larger UAVs are prioritized for assignment to ensure efficient resource utilization. If a fog node still has available resources after assigning UAVs ranked highest, the algorithm considers UAVs ranked lower (e.g., second choice) to balance resource allocation across the network. The rank-based approach ensures that high-priority fog nodes are fully utilized before moving to lower-priority nodes.

C. Complexity Analysis

The consensus-based fog node selection algorithm primarily involves identifying fog nodes within the communication range of UAVs and computing utility values for each UAV-fog node pair. This process entails nested iterations over UAVs and fog nodes, resulting in a complexity of $O(m \cdot n)$. Additionally, ranking fog nodes based on utility values introduces a sorting operation, contributing a complexity of $O(m \cdot n \log n)$. Furthermore, the calculation of CAP scores for all fog nodes has a complexity of $O(n)$, and sorting the fog nodes by their CAP scores adds a complexity of $O(n \log n)$. Subsequently, the Optimum Fog Node Resource Utilization algorithm, which is analogous to a best-fit bin packing problem, assigns UAVs to fog nodes while respecting resource constraints, incurring a complexity of $O(n \cdot m)$. Overall, combining these two algorithms, we have a combined computational complexity of $O(m \cdot n \log n)$, where m and n represent the number of UAVs and fog nodes, respectively. A detailed mathematical derivation of these complexities is provided in the supplementary file.

D. Theoretical Analysis

Theorem 1: The Con-Fog approach minimizes resource wastage and improves network performance as fog nodes increase within a fixed service area.

Proof: Consider a set of UAVs $U = \{u_1, u_2, \dots, u_m\}$ and fog nodes $F = \{f_1, f_2, \dots, f_n\}$ within a defined operational area, with each fog node f_j having a resource capacity of $\mathcal{R}_j^{f,\max}$. Each UAV u_i demands certain resources $\mathcal{R}_i^{u,\text{req}}$ for

effective operation. Define the average load $\mathcal{L}^d$ on a fog node as

$$\mathcal{L}^d = \frac{1}{n} \sum_{j=1}^n \left(\frac{\sum_{i \in S_j} \mathcal{R}_i^{u,\text{req}}}{\mathcal{R}_j^{f,\max}} \right) \quad (23)$$

where S_j represents the set of UAVs assigned to fog node f_j .

The Con-Fog approach systematically assigns UAVs to fog nodes, ensuring that no single node exceeds a predefined threshold, θ , for resource utilization

$$\forall j, \sum_{i \in S_j} \mathcal{R}_i^{u,\text{req}} \leq \theta \cdot \mathcal{R}_j^{f,\max}. \quad (24)$$

As the number of fog nodes (n) increases within a fixed area, and assuming a roughly uniform UAV distribution, the likelihood of each UAV being closer to a fog node increases. This can reduce the average distance between UAVs and fog nodes, thereby decreasing data transmission times, improving energy efficiency, and enhancing response times.

Next, consider the variance of the load distribution $\sigma^2(\mathcal{L}^d)$ across the fog nodes, which indicates the balance of load distribution

$$\sigma^2(\mathcal{L}^d) = \frac{1}{n} \sum_{j=1}^n \left(\left(\frac{\sum_{i \in S_j} \mathcal{R}_i^{u,\text{req}}}{\mathcal{R}_j^{f,\max}} \right) - \mathcal{L}^d \right)^2. \quad (25)$$

With an increase in n , the expected resource utilization per node decreases if UAV demands remain constant, and the distribution of load across nodes becomes more uniform, thus reducing $\sigma^2(\mathcal{L}^d)$.

Hence, by increasing the number of fog nodes within a fixed area, the Con-Fog approach effectively distributes UAVs among more nodes, optimizing resource utilization, minimizing the chance of any node's overutilization, and reducing the variance in resource distribution. This results in enhanced overall network performance and minimized resource wastage. ■

Lemma 1: With the increase in fog node density in the FU-Serve platform with a Con-Fog approach, the energy consumption per UAV decreases due to reduced transmission distance and time.

Proof: Given that the energy consumption for transmission $\mathcal{E}$ is directly proportional to the square of the transmission distance d and time t (i.e., $\mathcal{E} \propto d^2 \times t$), reducing d by assigning fog nodes closer to UAVs inherently decreases $\mathcal{E}$. As shown in Theorem 1, $\bar{d}$ decreases with increased fog node density, thus directly reducing $\mathcal{E}$ for UAV transmissions. ■

Proposition 1: In the Con-Fog approach, an increased density of fog nodes significantly reduces overall network latency, compared to the existing FU-Serve system, by ensuring that data processing occurs closer to data generation points.

Proof: Consider the network latency component $\mathcal{L}^t$ in a UAV network, which is predominantly influenced by the distance d between the UAVs and their respective data processing nodes (fog nodes). Latency $\mathcal{L}^t$ can be approximated by the equation

$$\mathcal{L}^t \approx k \cdot d \quad (26)$$

where k is a constant that includes factors like signal propagation speed and processing delays. In the Con-Fog approach, as the density of fog nodes n increases, the average distance d from any UAV to its nearest fog node decreases. Therefore, the average latency $\bar{\mathcal{L}}^t$ across the network is reduced

$$\bar{\mathcal{L}}^t = \frac{1}{n} \sum_{i=1}^n k \cdot \min_j d(u_i, f_j). \quad (27)$$

As m increases, $\min_j d(u_i, f_j)$ for each UAV u_i decreases, leading to a reduction in $\bar{\mathcal{L}}^t$. Therefore, with optimal assignment and increased density of fog nodes, the Con-Fog approach effectively minimizes the distance data must travel, thus reducing the overall system latency. ■

Proposition 2: Increasing the percentage of fog nodes in the FU-Serve platform with a Con-Fog approach significantly reduces the number of Unassigned UAVs, thereby enhancing the overall network utilization.

Proof: Consider a stochastic model for the UAV arrival rate, where $\lambda(t)$ varies over time, and a constant service rate per fog node μ . In an M/M/n queueing model with n fog nodes, the total service capacity is $n\mu$. The probability P_0 that no UAVs are waiting, which implies all are assigned, can still be approached by

$$P_0 = \left[\sum_{k=0}^n \frac{\left(\frac{\bar{\lambda}}{\mu}\right)^k}{k!} \right]^{-1} \quad (28)$$

where $\bar{\lambda}$ is the average arrival rate over a suitable period. This equation assumes the variability in $\lambda(t)$ can be averaged out over the considered time interval for analysis purposes.

As n increases, P_0 improves due to the increase in total service capacity $n\mu$, even as $\lambda(t)$ fluctuates. The system's utilization rate ρ , which takes into account the average arrival rate, is

$$\rho = \frac{\bar{\lambda}}{n\mu}. \quad (29)$$

Decreasing ρ as n increases suggests better capacity utilization, thus fewer UAVs remain unassigned despite the variability in arrival rates. Hence, an increased number of fog nodes enhances the system's robustness to fluctuations in UAV demand, leading to fewer unassigned UAVs and improved network utilization. ■

Theorem 2: The bin-packing strategy used for assigning UAVs to fog nodes in the FU-Serve platform minimizes the number of active fog nodes ($\sum y_j$) while ensuring no fog node's capacity is exceeded, thereby optimizing overall resource utilization.

Proof: Refer to the supplementary file. ■

Lemma 2: From Theorem 2, the best-fit decreasing algorithm minimizes the number of active fog nodes ($\sum y_j$).

Proof: Refer to the supplementary file. ■

V. PERFORMANCE EVALUATION

In this section, we present a comprehensive performance evaluation of the proposed Con-Fog approach to assess its

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Network Size	$1000 \times 1000m^2$
Number of UAVs	10, 20, 30, 40, 50
Number of Fog nodes	10%, 15%, 20%
Energy required to store 1 bit of data in a storage device	1nJ/bit
Energy spent by radio electronics	50nJ/bit
Transmitter amplifying free space coefficient	$10pJ/bit/m^2$
Transmitter amplifying multi-path coefficient	$0.001310pJ/bit/m^4$
Critical Data Percentage	25% and 50%
Propagation Model	Free Space and Multi-path Fading

effectiveness in optimally selecting fog nodes within the FU-Serve platform. To establish a baseline for comparison, we include the traditional UAVaaS platform [9], which typically relies on centralized cloud processing, and compare them with our FU-Serve platform [10] that integrates fog computing for enhanced responsiveness and reduced latency in UAV operations.

The evaluation focuses on several critical parameters, including transmission latency, energy consumption, and the number of unassigned UAVs, which are vital for determining the operational efficiency of the FU-Serve platform. By analyzing these parameters, we aim to demonstrate the enhanced capabilities of the Con-Fog approach over traditional UAVaaS and existing FU-Serve approaches, highlighting significant improvements in resource management and system responsiveness.

Our proposed fog node selection approach enhances the FU-Serve platform through several advancements. It integrates a consensus-driven decision-making process that accounts for a broad range of criteria, such as distance, link quality, and resource levels, through a voting mechanism among UAVs. This ensures that selections reflect both individual needs and collective preferences. The proposed approach's dynamic adaptability allows real-time adjustments to UAV assignments in response to operational changes, optimizing resource utilization and preventing inefficiencies. These advancements are supported by both theoretical analysis and empirical evidence, which demonstrate marked improvements in transmission times, energy efficiency, and overall resource management compared to existing systems.

Further confirming the effectiveness of the FU-Serve platform, this study highlighted how leveraging diverse UAV types—F-UAVs, H-UAVs, and N-UAVs—meets the intricate needs of the IoT ecosystem. However, integrating these heterogeneous UAV types presented significant challenges, particularly in achieving synchronized operations, which often led to delays and reduced data processing speeds during peak demands. Fluctuating fog node availability impacted system stability and response times, while the necessity to optimize resource allocation in a variable demand environment called for ongoing refinements to our algorithmic approach. These challenges underscore the need for continuous improvements in system design and operational strategies to fully harness the potential of UAV-enhanced fog computing within IoT frameworks.

A. Evaluation Metrics

We assess the performance of these systems using the following key metrics.

- 1) *Transmission Latency*: Measures the time taken for data to travel from the UAVs to the fog nodes, a critical factor in applications requiring real-time decision-making.
- 2) *Energy Consumption*: Evaluates the total energy used by UAVs and fog nodes during operations, which is essential for sustainable long-term deployment.
- 3) *Number of Unassigned UAVs*: Indicates the efficiency of UAV allocation to fog nodes, with lower numbers suggesting better resource utilization and system configuration.

The subsequent subsections will detail the methodologies employed in the simulation, present the results, and discuss the implications of these findings in the context of enhancing IoT applications with UAVs. By providing a clear comparison with established systems, this evaluation aims to substantiate the benefits of the Con-Fog approach, emphasizing its potential to revolutionize UAV service delivery in various sectors.

B. Simulation Setup and Parameters

In our simulation, we consider both static and dynamic fog nodes to closely mimic real-world scenarios. The specifications of the simulation parameters, including characteristics of UAVs and configurations of fog nodes, are detailed in Table II. The simulation area is divided into grids, each containing a static fog node at its center to optimize local data processing and responsiveness. To explore the dynamics of UAVs, we implemented varied movement modes in our simulation, reflecting realistic UAV operational behaviors as follows.

- 1) *Predefined Path*: UAVs follow predefined paths to perform routine surveillance and data collection.
- 2) *Event-Driven*: UAVs are redirected in response to specific environmental events or data demands, simulating emergency response scenarios.
- 3) *Random Mobility*: UAVs move randomly within the simulation area to mimic ad hoc data collection tasks.

The utility of our proposed Con-Fog approach in dynamic environments is assessed using the algorithm proposed in [10], which selects UAVs as dynamic fog nodes based on their energy, average load, and proximity to high data demand areas. The fleet of UAVs are as follows.

- 1) *F-UAVs*: Equipped exclusively with fog computing capabilities, F-UAVs play a key role in FU-Serve by providing onboard processing power for real-time tasks.
- 2) *H-UAVs*: These UAVs have advanced sensors and limited fog computing resources. This fleet assesses the ability of the fleet to handle moderately complex tasks, such as high-resolution imaging and on-the-spot data processing.
- 3) *N-UAVs*: Equipped with basic sensors that are required for serving applications. An N-UAV is responsible for performing traditional data collection tasks.

To cater to the demands of time-critical applications, we designate 25% and 50% of the data as critical, necessitating expedited processing to meet strict response time requirements. This data categorization enables a thorough assessment

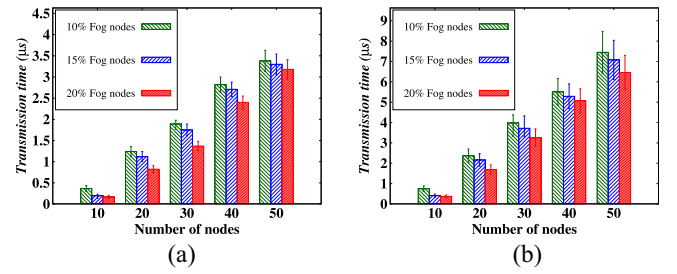


Fig. 2. Comparison of transmission times with varying percentages of fog nodes. (a) shows the scenario with 25% critical data and (b) shows the scenario with 50% critical data. (a) Critical data = 25%. (b) Critical data = 50%.

of the performance of Con-Fog under varying levels of time-critical scenarios.

C. Transmission Time Analysis With Critical Data

This section evaluates the effectiveness of the Con-Fog approach in reducing transmission times, with an emphasis on the impact of increasing fog node density and the strategic assignment of UAVs to these nodes. As we increment the number of UAVs from 10 to 50, a natural increase in network load and complexity occurs, leading to longer transmission times. However, an increase in the density of fog nodes (10%, 15%, and 20%) significantly counters this effect, particularly when 50% of the transmitted data is critical and necessitates swift processing. The detailed results of this analysis are depicted in Fig. 2(a) and (b), which illustrate how transmission times initially escalate with an increase in the number of UAVs but are substantially reduced as fog node density increases. The improvement emphasizes the utility of localized data processing, especially under conditions of heavy data load and critical operational scenarios, as evidenced in tests with both 25% and 50% critical data.

We consider 25% and 50% critical data ratios in our simulation to reflect varying operational urgencies within real-world IoT applications. The 25% ratio simulates scenarios where a significant, yet not predominant, portion of data requires prompt processing, typical of routine monitoring tasks in urban planning or agriculture. Conversely, 50% ratio tests the resilience of the system under high urgency conditions, such as disaster response or large-scale event monitoring, where rapid processing of a significant volume of critical data is crucial for effective management and response. These ratios allow us to comprehensively evaluate the robustness and adaptability of Fu-Serve, demonstrating its capability to handle varying levels of data criticality.

Further granularity is provided in Fig. 3(a)–(c), which compares transmission times at various fog node densities for scenarios with 25% critical data. Similarly, Fig. 3(b), (d), and (f) details the impact on scenarios where 50% of the data is critical. We observed that in scenarios with increased fog node density, such as in Fig. 3(a) and (b), transmission time increased due to resource contention in H-UAVs and F-UAVs. As fog density increases, the cumulative background load on fog nodes rises. Simultaneously, competing requests from H-UAVs and N-UAVs for shared resources create contention

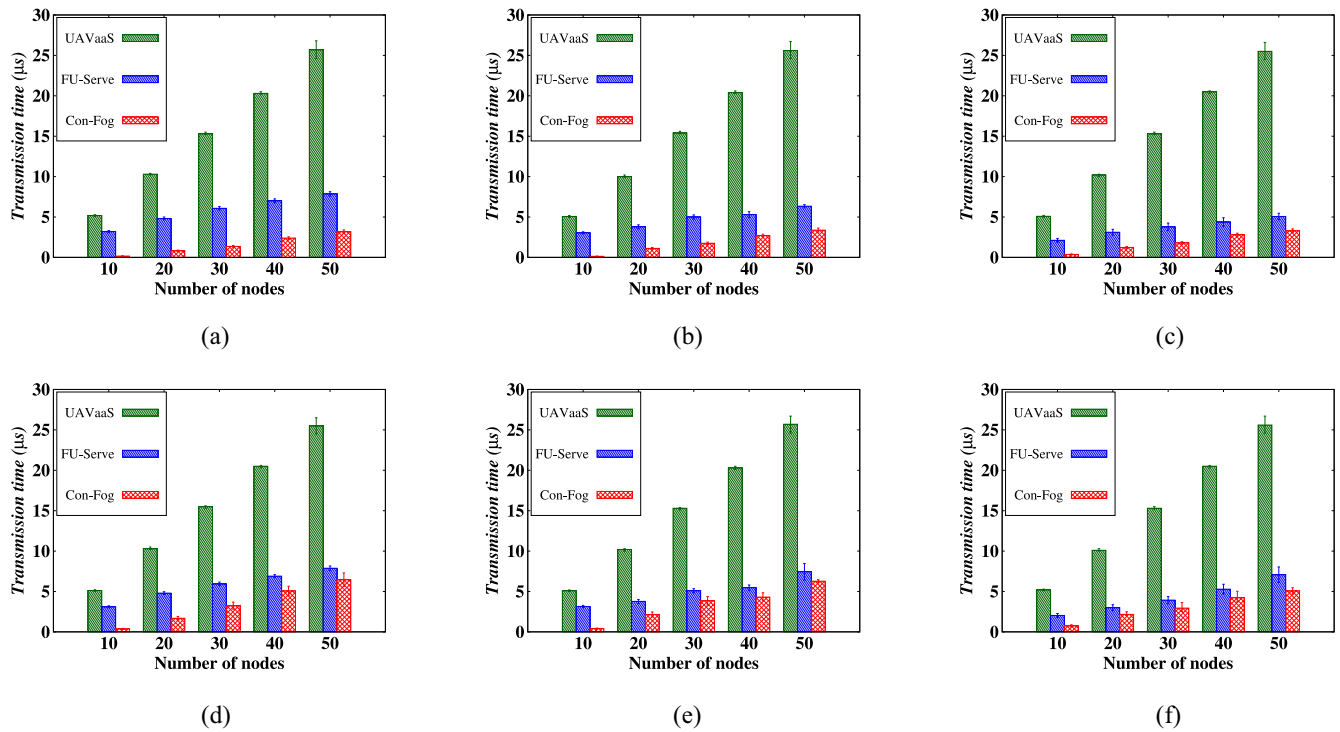


Fig. 3. Comparative analysis of transmission times under different configurations of fog node density and critical data percentages. These figures demonstrate the impact of increasing fog node density on reducing transmission times in scenarios with 25% and 50% critical data. (a) Transmission time with 10% fog nodes and 25% critical data. (b) Transmission time with 15% fog nodes and 25% critical data. (c) Transmission time with 20% fog nodes and 25% critical data. (d) Transmission time with 10% fog nodes and 50% critical data. (e) Transmission time with 15% fog nodes and 50% critical data. (f) Transmission time with 20% fog nodes and 50% critical data.

between background tasks and dynamic workloads, leading to queueing delays. For instance, with 20 nodes, a higher density of F-UAVs and H-UAVs results in slight delays due to multitasking across multiple applications. Nevertheless, the transmission time remains lower than that of both the UAVaaS and traditional FU-Serve platforms. Additionally, we notice the change in transmission times between Figs. 2(b) and 3(f) (both at 20% fog node density) arises from dynamic network conditions, UAV positions relative to fog nodes, and task-specific processing demands. In all instances, the Con-Fog approach consistently outperforms traditional UAVaaS and existing FU-Serve systems by ensuring optimal selection of fog nodes in proximity to UAVs, thereby minimizing the distance data needs to travel and maximizing the efficient use of computational and storage resources at each node. This approach effectively mitigates bottlenecks and accelerates data processing across the network.

D. Energy Consumption Trends Analysis With Critical Data

This analysis evaluates energy consumption trends on the FU-Serve platform using the Con-Fog approach, depicted in Fig. 4(a) and (b). These figures illustrate the remaining energy in fog nodes under scenarios with 25% and 50% critical data. The data shows a decrease in leftover energy in fog nodes as the number of UAVs increases, indicating higher energy usage due to enhanced data processing and communication tasks. As the density of fog nodes increases from 10%, 15% to 20%, there is a notable improvement in energy conservation. This

distribution helps avoid overloading individual nodes, thus enhancing overall energy efficiency. In scenarios with 50% critical data, fog nodes have slightly less remaining energy compared to the 25% critical data scenario, reflecting the greater energy demands of processing time-critical data.

Comparative analysis in Figs. 5 and 6 shows the Con-Fog approach's superior performance over the existing FU-Serve system. The proposed approach maintains higher residual energy levels, indicating more effective energy management strategies, particularly under higher critical data volumes. This performance advantage emphasizes the efficacy of the Con-Fog approach in managing increased operational demands without sacrificing energy efficiency, thus improving the sustainability and scalability of UAV networks. The strategic fog node selection optimizes energy consumption, extending the operational duration of UAVs and reducing both environmental impact and operational costs.

E. Unassigned UAVs: Existing Versus Con-Fog Approach

The analysis of unassigned UAVs within the FU-Serve platform highlights the efficiency of UAV resource utilization, which is crucial for applications requiring rapid response times, such as environmental monitoring and emergency services. Minimizing the number of unassigned UAVs is essential to maximize throughput and responsiveness. Figs. 7 and 8 illustrate the comparative performance between the existing system and the Con-Fog approach. These figures show a decrease in the number of unassigned UAVs as the percentage

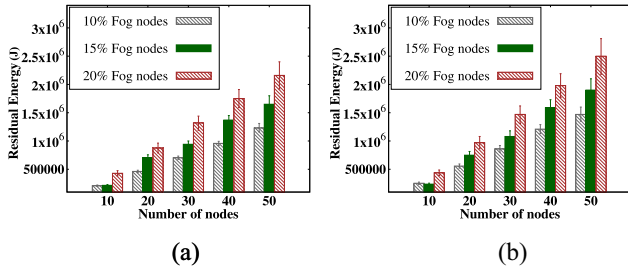


Fig. 4. Comparison of residual energy at fog nodes with varying percentages of fog nodes. (a) shows the scenario with 25% critical data and (b) shows the scenario with 50% critical data. (a) Critical data = 25%. (b) Critical data = 50%.

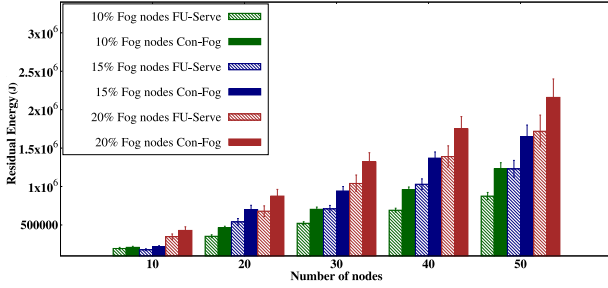


Fig. 5. Energy comparison for 25% critical data between the existing FU-Serve and proposed Con-Fog approach.

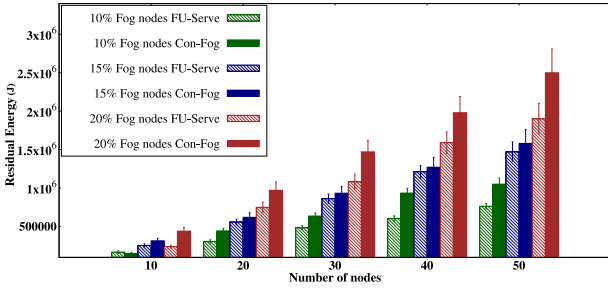


Fig. 6. Energy comparison for 50% critical data between the existing FU-Serve and proposed Con-Fog approach.

of fog nodes increases in scenarios with 25% and 50% critical data. The Con-Fog approach optimizes UAV allocation by effectively distributing UAV load across the network's fog nodes, enhancing distributed processing capabilities. This optimization results in superior load distribution, significantly reducing the number of unassigned UAVs and ensuring that all fog nodes are utilized to their full potential. Such improvements are crucial in high-stakes environments where efficient resource use is critical. By enhancing operational efficiency and reducing unassigned UAVs, the Con-Fog approach proves to be a robust solution for improving the responsiveness and effectiveness of FU-Serve services in various sectors requiring quick and reliable UAV operations.

VI. CONCLUSION

In this article, we have thoroughly analyzed the existing fog node selection approaches within the FU-Serve platform, particularly in the context of IoT applications, and compared them to our newly proposed Con-Fog approach. Our method

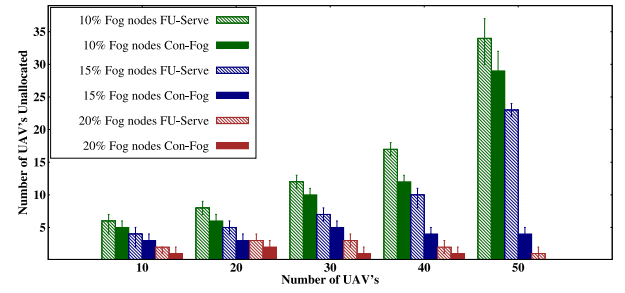


Fig. 7. Comparison of unassigned UAVs between existing and proposed methods with 25% critical data.

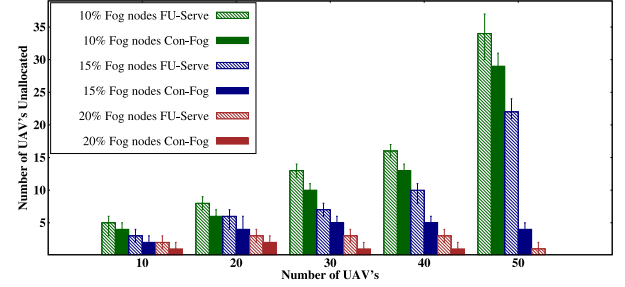


Fig. 8. Comparison of unassigned UAVs between existing and proposed approach with 50% critical data.

is specifically designed to select the optimal fog node for each UAV, thereby optimizing load balance across the network by incorporating UAV preferences into the selection process. We began with a detailed introduction to the inherent challenges of the FU-Serve platform, particularly the critical need for efficient fog node selection to mitigate issues related to latency, energy consumption, and the diverse demands of IoT applications. Our Con-Fog approach, grounded in Social Choice theory, introduces a structured, consensus-driven method for fog node selection that effectively integrates individual UAV preferences into collective decision-making. This method has significantly improved the performance of the FU-Serve platform, as demonstrated by substantial reductions in transmission times and enhancements in energy efficiency observed in various simulations. The results show that the Con-Fog approach surpasses existing systems by reducing transmission times, improving energy management, and reducing the number of unassigned UAVs, particularly under the high data traffic and dynamic conditions typical of IoT environments. This effective management of fog node selection and utilization is crucial for supporting efficient UAV operations and meeting the stringent demands of IoT applications. In the future, we plan to incorporate AI and machine learning techniques to predict traffic patterns, UAV behaviors, and fog node performance, aiming to optimize resource allocation strategies further and proactively manage potential bottlenecks within IoT-driven ecosystems.

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